Supplement to "Cross-Cluster Weighted Forests"

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1 Considering a single covariate

In this section, we consider datasets with a single covariate, so that we can visualize the differences between the true outcome model and the estimated outcome models from the *Merged* and *Ensemble* as a function of the input covariate. We consider a cubic outcome model and the gaussian cluster-generating framework with two clusters.

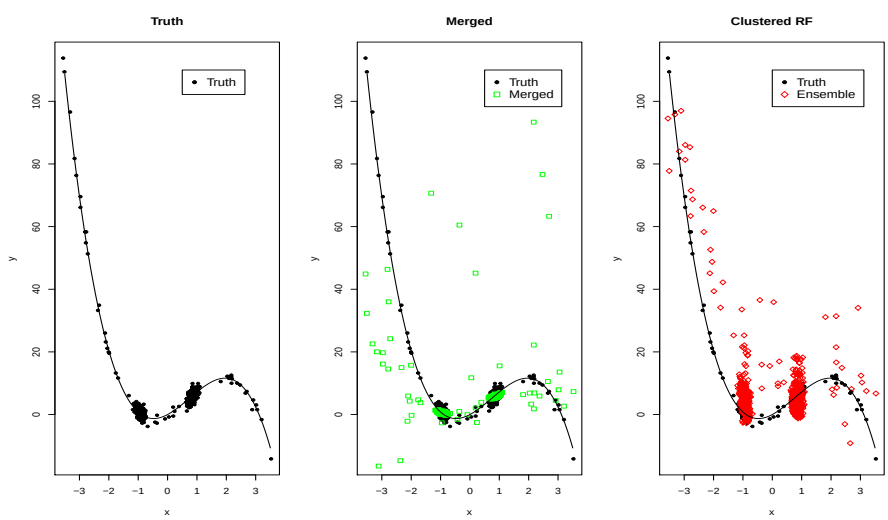

Figure S1: Cubic outcome model: $y = 5x_1 + 4x_1^2 - 1.8x_1^3$. The single covariate x is simulated as a mixture model of two gaussian clusters. The leftmost panel displays the covariate data and the true outcomes, as well as the underlying cubic model. The center and right panels displays the predictions of the *Merged* and *Cluster* learners respectively, as well as the underlying true outcome values and model. The *Cluster* approach has an approximately 15% gain in predictive performance over the *Merged*.

The true model is visualized by the black line in all of the panels and the true outcome values corresponding to the generated covariate data is displayed by the black points. The predictions by the *Merged* and *Ensemble* are depicted in green and red respectively. In comparing the prediction accuracy of the two learning approaches, we find that in this instance,

the *Ensemble* approach outperforms the *Merged* by 15%. Figure S1 allows us to visualize where these improvements occur; while the *Merged* is good at learning the true covariate-outcome at the densest regions of the covariate space (in this case, corresponding to the centers of the two clusters), the *Ensemble* is able to additionally more closely learn the true model at the tails of the covariate distribution. By focusing each forest on a single cluster, the *Ensemble* method is able to focus more on all areas of the covariate space, while the *Merged* concentrates on those with most representation within the training set.

2 Weighting trees within the *Ensemble*

In this section, we explore the effects of computing stacking weights on the tree-level to form the final ensemble, instead of simply weighting each cluster-specific forest. In this framework, individual forests are still trained on each data-partition found by a clustering algorithm, random subsetting, or the true simulated clusters; however, now each tree within the forest is extracted and stacking weights are computed, upweighting those trees that show better cross-cluster prediction ability. This framework has been explored extensively in the multi-study setting in [1]; we now determine whether it produces improvements for single training datasets as well.

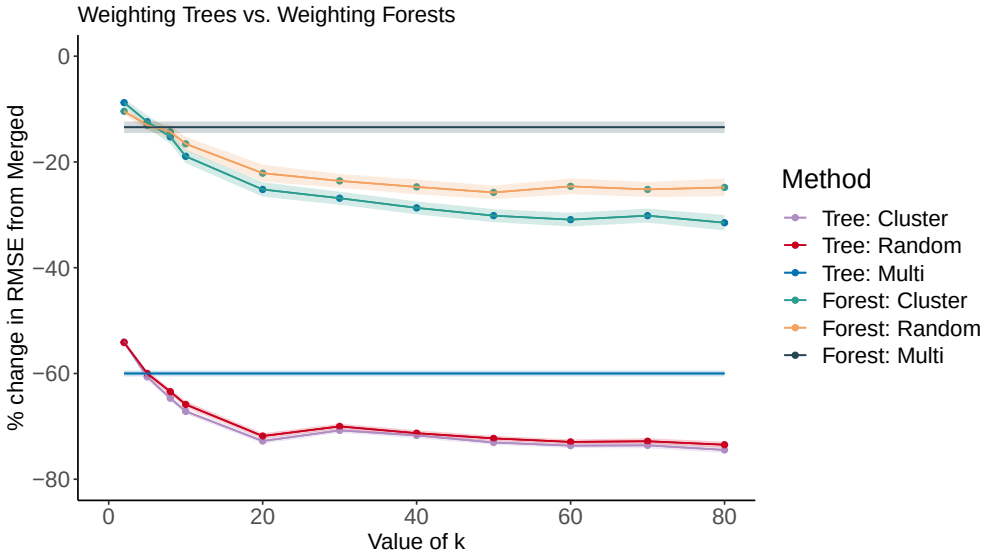

Figure S2: Percent change in average RMSE of ensembling approaches (color labeled) compared to the Merged comparing weighting trees vs. weighting forests using the non-gaussian cluster-generating framework and a quadratic outcome model.

Figure S2 displays the performance of ensembles in which either the individual trees or forests are weighted through stacked regression, using the three main ensembling approaches discussed throughout the paper on the non-gaussian simulation setup with a quadratic outcome model. We observe that as in the multi-study setting discussed in [1], weighting trees produces significant improvements upon weighting forests. However, among the tree-weighting approaches, there is no discernible difference between the *Cluster* and the *Random*

methods, indicating that allowing for up-weighting of individual trees removes the need for optimally-chosen data partitions and simply requires that each partition reduce the total covariate space sufficiently. Overall, these results suggest that weighting trees instead of forests within this paradigm can further increase prediction accuracy, and should be further explored.

References

[1] Maya Ramchandran, Prasad Patil, and Giovanni Parmigiani. Tree-weighting for multi-study ensemble learners. *Pacific Symposium on Biocomputing*, 25:451–462, 2020.